Section 6.1 Games, invariants & state strategies

博弈、不变量与状态策略

Method 1 / 方法 1 [LO01-M01]

An invariant proves a necessary restriction; a construction or a legal strategy proves sufficiency. When a state changes by folding, cancellation, or a custom operation, record the information that survives that change.

不变量证明必要限制，构造或合法策略证明充分性。面对折叠、约分或自定义运算，应记录在变化中仍有效的信息。

Prove losing residues by induction on legal moves

按合法移动归纳证明必败余数

Procedure

1. Define a losing position as one whose every legal move is winning; a winning position needs one legal move to a losing state.

2. Guess a residue pattern only after computing enough small positions.

3. Prove both directions of the residue rule, checking that the proposed move is available at small pile sizes.

4. Count or apply the resulting strategy, with zero treated as a losing terminal state under normal play.

通用步骤

1. 必败状态的所有合法后继都必胜；必胜状态只需存在一个合法必败后继。

2. 先算足够多的小状态，再猜测余数模式。

3. 证明余数规则的两个方向，并检查小堆中所提移动是否合法。

4. 由规则计数或执行策略；正常取尽获胜规则下，零为必败终态。

Section 6.1 Games, invariants & state strategies

博弈、不变量与状态策略

Method 1 / 方法 1 [LO01-M01] (continued / 续)

> Example. Players alternately remove $1, 3$, or 4 tokens, with the last removal winning. Classify the losing pile sizes.

> 例题：两人轮流取走一、三或四枚棋子，取尽者胜，分类所有必败堆大小。

Worked solution

1. Direct computation for 0 through 6 gives losing sizes $0, 2$ and all other sizes winning.

2. From residues $0, 2 \pmod 7$, subtracting a legal $1, 3, 4$ always leads to residues $1, 3, 4, 5, 6$, which are outside the proposed losing class.

3. From residues $1, 3, 4, 5, 6$, respectively subtract $1, 1, 4, 3, 4$ to reach residue 0 or 2. These moves are legal at the least positive representatives. Induction proves exactly $n \equiv 0, 2 \pmod 7$ are losing.

例题解析

1. 直接计算零至六，必败大小为零、二，其余必胜。

2. 从模七余零或二的状态，合法减去一、三、四后，都到余一、三、四、五、六，不在拟定必败类中。

3. 从余一、三、四、五、六分别减一、一、四、三、四，会到余零或二，且在最小正代表处也合法。归纳得必败状态恰为 $n \equiv 0, 2 \pmod 7$。

A repeating numerical table suggests a strategy; only the two legal-move implications prove it.

数值表重复只提示策略，必须证明合法移动的两个方向才算完成。

Section 6.1 Games, invariants & state strategies

博弈、不变量与状态策略

Method 1 / 方法 1 [LO01-M01] (continued / 续)

Practice / 配套练习: AIME-2024-I-Q03 (WB p. 797)

Section 6.1 Games, invariants & state strategies

博弈、不变量与状态策略

Method 2 / 方法 2 [LO01-M02]

Propagate forced colors, then construct below the threshold

传播被迫颜色并在门槛以下构造

Procedure

1. Assume a forbidden monochromatic relation is absent and fix one initial color by symmetry.

2. Propagate forced opposite colors through the relation.

3. Find an element whose two possible colors both cause a contradiction.

4. For a sharp threshold, provide an explicit coloring below it and check every same-color relation.

通用步骤

1. 假设不存在禁止的同色关系，并利用颜色对称固定起始颜色。

2. 沿关系传播被迫的相反颜色。

3. 找出一个无论染哪色都矛盾的元素。

4. 若要求精确门槛，必须给出门槛以下的明确染色并核对全部同色关系。

> Example. Find the least m such that every two-coloring of $\{1,\dots,m\}$ has same-colored a,b,c with $a+b=c$, allowing $a=b$.

> 例题：求最小 m，使 $\{1,\dots,m\}$ 的任意两色染色都存在同色 a,b,c 满足 $a+b=c$，允许 $a=b$。

Section 6.1 Games, invariants & state strategies

博弈、不变量与状态策略

Method 2 / 方法 2 [LO01-M02] (continued / 续)

Worked solution

1. At $m = 5$, color one red. Avoiding $1 + 1 = 2$ forces two blue; avoiding $2 + 2 = 4$ forces four red; avoiding $1 + 4 = 5$ forces five blue.

2. If three is red, $1 + 3 = 4$ is monochromatic; if blue, $2 + 3 = 5$ is monochromatic. Hence five always forces the relation.

3. For $m = 4$, color $\{1, 4\}$ red and $\{2, 3\}$ blue. A direct check of sums within each color finds none in that same color. Restricting this coloring covers all smaller m, so the least is five.

例题解析

1. 当 $m = 5$，不妨一为红。由 $1 + 1 = 2$，二须蓝；由 $2 + 2 = 4$，四须红；由 $1 + 4 = 5$，五须蓝。

2. 若三红，$1 + 3 = 4$ 同色；若三蓝，$2 + 3 = 5$ 同色，因此五必然产生目标关系。

3. 当 $m = 4$，把一、四染红，二、三染蓝，逐一检查同色和均不落在本色中；更小范围可限制该染色。因此最小值为五。

A contradiction proves an upper bound; an explicit avoiding coloring proves the matching lower bound.

矛盾只证明上界，明确的避免染色才能证明相应下界。

Practice / 配套练习: AIME-2010-I-Q12 (WB p. 798)

Section 6.1 Games, invariants & state strategies

博弈、不变量与状态策略

Method 3 / 方法 3 [LO01-M03]

Pair a modular invariant with a forward reachability certificate

将同余不变量与正向可达构造配对

Procedure

1. Find linear combinations of coordinates preserved modulo small integers by every allowed move.

2. Treat these as necessary conditions only.

3. Build nonnegative combinations of legal moves that generate every point satisfying the conditions; integer combinations with negative move counts are insufficient.

4. When counting, transform the region and retain any parity compatibility between new coordinates.

通用步骤

1. 寻找每种合法移动都保持的坐标线性组合及小模数。

2. 先把它们视为必要条件。

3. 以合法移动的非负次数组合构造全部满足条件的点；允许负次数的整数线性组合还不够。

4. 计数时转换区域，并保留新坐标间的奇偶兼容条件。

> Example. From the origin, allowed moves are $(2,0),(0,2),(-1,-1)$. Characterize all reachable integer points.

> 例题：从原点出发，可走 $(2,0),(0,2),(-1,-1)$，刻画全部可达整数点。

Section 6.1 Games, invariants & state strategies

博弈、不变量与状态策略

Method 3 / 方法 3 [LO01-M03] (continued / 续)

Worked solution

1. Each move preserves evenness of $x-y$, so reachable coordinates must have the same parity.

2. For any such target (x,y), choose an integer $c \geq \max(0,-x,-y)$ with the common parity of x,y. Then $a=(x+c)/2, b=(y+c)/2$ are nonnegative integers.

3. The legal sequence using the moves a,b,c times reaches $(2a-c,2b-c)=(x,y)$. Thus equal coordinate parity is both necessary and sufficient.

例题解析

1. 每步均保持 $x-y$ 为偶数，所以可达点两坐标须同奇偶。

2. 对任一这样的目标，选与两坐标同奇偶且 $c \geq \max(0,-x,-y)$ 的整数，令 $a=(x+c)/2, b=(y+c)/2$，均为非负整数。

3. 三种移动分别走 a,b,c 次，就到达 $(2a-c,2b-c)=(x,y)$，故同奇偶既必要又充分。

A lattice generated algebraically by moves can be larger than the forward-reachable semigroup; prove the nonnegative construction.

移动所生成的整数格可能大于正向可达半群，必须证明非负次数构造。

Practice / 配套练习: AIME-2012-I-Q11 (WB p. 799)

Section 6.1 Games, invariants & state strategies

博弈、不变量与状态策略

Method 4 / 方法 4 [LO01-M04]

Search for a conserved algebraic expression before iterating

迭代前先寻找守恒代数式

Procedure

1. Write the transformation on a complete tuple and compare the same expression before and after it.

2. Clear denominators or factor the difference to prove exact invariance.

3. Use the invariant to reduce a nonlinear recurrence to a linear relation or a fixed algebraic curve.

4. Restore positivity and monotonic bounds before extracting floors or extrema.

通用步骤

1. 把变换写在完整元组上，比较变换前后的同一个表达式。

2. 清分母或将差因式分解，证明精确不变。

3. 利用不变量把非线性递推化为线性关系或固定代数曲线。

4. 求整数部分或最值前，恢复正性及单调界。

> Example. Show that replacing c by $3ab-c$ preserves $a^2+b^2+c^2-3abc$. Apply it to $(1,1,1)$.

> 例题：证明将 c 换成 $3ab-c$ 保持 $a^2+b^2+c^2-3abc$ 不变，并应用于 $(1,1,1)$。

Section 6.1 Games, invariants & state strategies

博弈、不变量与状态策略

Method 4 / 方法 4 [LO01-M04] (continued / 续)

Worked solution

1. Write $c' = 3ab - c$. Then $(c')^2 - 3abc' = 9a^2b^2 - 6abc + c^2 - 9a^2b^2 + 3abc = c^2 - 3abc$.

2. The other terms $a^2 + b^2$ are unchanged, proving the invariant.

3. The initial value is zero. The replacement gives $(1, 1, 2)$, and $1 + 1 + 4 - 6 = 0$ verifies the same algebraic curve.

例题解析

1. 令 $c' = 3ab - c$，展开得 $(c')^2 - 3abc' = c^2 - 3abc$。

2. 其余两项 $a^2 + b^2$ 不变，所以表达式守恒。

3. 起始值为零，变换得到 $(1, 1, 2)$，且 $1 + 1 + 4 - 6 = 0$，仍位于同一代数曲面。

An invariant is useful only after its domain and any needed inequality bounds are checked.

只有核对定义域及所需不等式界后，不变量才足以支持结论。

If $ac = b^2 + C$ and $bd = c^2 + C$, then $c(a + c) = b(b + d)$, hence $(a + c)/b = (b + d)/c$ for nonzero b, c. Thus the adjacent ratio $K = (a_{n-1} + a_{n+1})/a_n$ is constant and the recurrence becomes $a_{n+1} = Ka_n - a_{n-1}$. The original identity also gives $a_{n+1}^2 + a_n^2 = Ka_na_{n+1} - C$.

若 $ac = b^2 + C, bd = c^2 + C$，则 $c(a + c) = b(b + d)$，非零 b, c 下有 $(a + c)/b = (b + d)/c$。因此相邻比值 $K = (a_{n-1} + a_{n+1})/a_n$ 为常数，递推化为 $a_{n+1} = Ka_n - a_{n-1}$；原式还给出 $a_{n+1}^2 + a_n^2 = Ka_na_{n+1} - C$。

Practice / 配套练习: AIME-2007-I-Q14 (WB p. 158)

Section 6.1 Games, invariants & state strategies

博弈、不变量与状态策略

Method 5 / 方法 5 [LO01-M05]

Track an invariant through reduction of rational pairs

在分数约分中跟踪不变量

Procedure

1. Use the reduced numerator and denominator as the state, because the next rule depends on them.

2. Find a linear determinant invariant before reduction; after dividing by a gcd, that invariant divides by the same factor.

3. For the shift $(m, n) \mapsto (m + r, n + r + 1)$, put $d = n - m, I = (r + 1)m - rn$. The cancellation factor is $\gcd(d + 1, I)$.

4. Use the new difference and invariant together to prove whether a target shape can recur.

通用步骤

1. 将既约分子分母作为状态，因为下一步规则依赖它们。

2. 先找约分前的线性行列式不变量；除以最大公因数后，它也除以同一因子。

3. 若平移为 $(m, n) \mapsto (m + r, n + r + 1)$，令 $d = n - m, I = (r + 1)m - rn$，约分因子为 $\gcd(d + 1, I)$。

4. 同时跟踪新差值与不变量，以证明某个目标形式能否再次出现。

> Example. Start with $5/7$. Repeatedly add two to its reduced numerator and three to its reduced denominator, then reduce. Can a term equal $t/(t + 1)$ for a positive integer t?

> 例题：从 $5/7$ 出发，反复给既约分子加二、分母加三，再约分，能否出现正整数 t 的 $t/(t + 1)$?

Section 6.1 Games, invariants & state strategies

博弈、不变量与状态策略

Method 5 / 方法 5 [LO01-M05] (continued / 续)

Worked solution

1. Here $I = 3m - 2n$ starts at $15 - 14 = 1$ and is unchanged before reduction.

2. The cancellation factor is $\gcd(d+1, 1) = 1$, so no reduction ever occurs and d increases by one each time.

3. Initially $d = 2$, hence always $d \geq 2$. A reduced fraction $t/(t+1)$ has difference one, so it can never occur.

例题解析

1. 此处 $I = 3m - 2n$ 初值为一，未约分前保持不变。

2. 约分因子为 $\gcd(d+1, 1) = 1$，所以永不约分，差 d 每步增加一。

3. 初始差为二，始终至少二，而 $t/(t+1)$ 已既约且差为一，因此不可能出现。

An unreduced update and a reduced update are different dynamical systems.

未约分与约分后的递推是不同的动力系统。

Practice / 配套练习: AIME-2021-I-Q10 (WB p. 491)

Section 6.1 Games, invariants & state strategies

博弈、不变量与状态策略

Method 6 / 方法 6 [LO01-M06]

Track both position and layer orientation in repeated folds

反复折叠时同时跟踪位置与层次反向

Procedure

1. Record current strip length L, stack thickness H, horizontal unit position p, and bottom-up layer b.

2. A square in the retained half keeps its state. A right-half square folded onto the left changes to $p' = L + 1 - p, b' = 2H + 1 - b$.

3. After the fold, halve L and double H, then repeat.

4. For an entire stack, append the reversed moving-layer list above the retained-layer list.

通用步骤

1. 记录当前条长 L、厚度 H、水平位置 p 及自下而上层号 b。

2. 保留半区的方格状态不变；右半向左盖上时，$p' = L + 1 - p, b' = 2H + 1 - b$。

3. 每次折后 L 减半、H 加倍，再重复。

4. 若跟踪整叠顺序，应把移动侧层表反向后接在保留侧层表之上。

> Example. An eight-square strip is labeled 1 through 8. Fold the right half over the left three times. Give the final stack from bottom to top.

> 例题：八格纸条从左到右标一至八，连续三次将右半盖到左半，求最终自下而上层序。

Section 6.1 Games, invariants & state strategies

博弈、不变量与状态策略

Method 6 / 方法 6 [LO01-M06] (continued / 续)

Worked solution

1. After the first fold, the four vertical stacks from left to right are $[1,8]$, $[2,7]$, $[3,6]$, $[4,5]$.

2. After the next fold, they are $[1,8,5,4]$ and $[2,7,6,3]$, reversing each moved stack.

3. The final fold gives $[1,8,5,4,3,6,7,2]$. Reversing horizontal order alone would miss the reversed layer orientation.

例题解析

1. 第一折后，从左到右四叠为 $[1,8]$, $[2,7]$, $[3,6]$, $[4,5]$。

2. 第二折时反向移动层表，得到 $[1,8,5,4]$ 及 $[2,7,6,3]$。

3. 最后得到 $[1,8,5,4,3,6,7,2]$。只反转水平位置而不反转层次会出错。

Use bottom-up or top-down layers consistently; changing conventions mid-process reverses the answer.

必须始终统一用自下而上或自上而下层号，途中换约定会反转答案。

Practice / 配套练习: AIME-2004-II-Q15 (WB p. 390)

Section 6.1 Games, invariants & state strategies

博弈、不变量与状态策略

Method 7 / 方法 7 [LO01-M07]

Use ordinary prefix parity to control a custom operation

用普通前缀奇偶控制自定义运算

Procedure

1. Compare the operation modulo two; addition and subtraction have the same parity effect.

2. Prove that the running-value parity equals the ordinary prefix-sum parity.

3. Mark exactly which terms are subtracted; the final value equals the ordinary sum minus twice their total.

4. For counting, keep both total size and running value as states and extend by every allowed positive part.

通用步骤

1. 先模二比较运算，加与减对奇偶的影响相同。

2. 证明当前运算值与普通前缀和同奇偶。

3. 标出恰被减去的项，最终值等于普通总和减去这些项总和的两倍。

4. 计数时同时保存总量及当前值，并用每个允许正部件作扩展。

> Example. Define $a \circ b = a - b$ if a is odd and b even, and $a + b$ otherwise. Evaluate the left-associated composition $2 \circ 1 \circ 4 \circ 3$ using prefix parity.

> 例题：定义 $a \circ b$：若 a 奇且 b 偶则取 $a - b$，否则取 $a + b$。用前缀奇偶求左结合的 $2 \circ 1 \circ 4 \circ 3$。

Section 6.1 Games, invariants & state strategies

博弈、不变量与状态策略

Method 7 / 方法 7 [LO01-M07] (continued / 续)

Worked solution

1. The ordinary prefixes before the later terms are $2, 3, 7$, with parities even, odd, odd.

2. Only the even term four following the odd prefix three is subtracted. The other terms are added.

3. Thus the result is $(2 + 1 + 4 + 3) - 2 \cdot 4 = 2$. Direct left evaluation $2 \circ 1 = 3, 3 \circ 4 = -1, -1 \circ 3 = 2$ confirms it.

例题解析

1. 后三项之前的普通前缀分别为二、三、七，奇偶为偶、奇、奇。

2. 只有跟在奇前缀三后的偶数四被减去，其余均相加。

3. 所以结果为 $10 - 2 \cdot 4 = 2$；直接左结合计算依次为三、负一、二，吻合。

The operation is left-associated and generally not associative; parentheses are part of the state rule.

运算按左结合进行，一般不满足结合律，括号顺序属于状态规则的一部分。

Practice / 配套练习: AIME-2026-II-Q14 (WB p. 689)